- Gen. fn. : For X, non-neg int valued r.v.

$$P_X(s) = \sum_{n=0}^{\infty} p_n s^n = \sum_{n=0}^{\infty} P(X=n) s^n$$

$$= E(s^X), \quad |s|<1.$$

- X, Y indept,

$$P_{X+Y}(s) = E(s^{X+Y}) = E(s^X s^Y)$$

$$= E(s^X) E(s^Y). \quad (\Rightarrow \text{ind.})$$

$$= P_X(s) P_Y(s)$$

$$\mathbb{P}(X+Y=n) = \sum_{k=0}^{n} \mathbb{P}(X=k, X+Y=n)$$

$$= \mathbb{P}(X=k, Y=n-k)$$

$$= P(X=k)\,P(Y=n-k)$$

$$= \sum_{k=0}^{n} P(X=k)\,P(Y=n-k)$$

$$= \sum_{k=0}^{n} \mathbb{P}(Y=k)\,P(X=n-k)$$

$\{a_n\}, \{b_n\}$

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}$$

$\{c_n\}$ is convolution of $\{a_n\}$ and $\{b_n\}$

<u>Thm</u> :- Let $A(s)$ & $B(s)$ be the gen fns of $\{a_n\}$ and $\{b_n\}$ resp. Assume that the rad. of conv. is at least R for both $A(s)$ & $B(s)$. Then, the conv. seq^n $\{c_n\}$, has a rad. of conv at least R and the gen. fn is given by $A(s)B(s)$.

Pf:- Fix $\epsilon > 0$, then

$$\limsup_n |a_n|^{1/n} \leq \frac{1}{R} + \epsilon$$

$\exists N, \forall n \geq N$

$$|a_n|^{1/n} \leq \frac{1}{R} + \epsilon$$

$$\Rightarrow |a_n| \leq \left(\frac{1}{R} + \epsilon\right)^n$$

$$|a_n| \leq A\left(\frac{1}{R} + \epsilon\right)^n$$ where $A, B > 0$ is a const

$$|b_n| \leq B\left(\frac{1}{R} + \epsilon\right)^n$$

$$|c_n| = \left|\sum_{k=0}^{n} a_k b_{n-k}\right| \leq \sum_{0}^{n} |a_k|\,|b_{n-k}|$$

$$\leq \sum^{n} A\left(\frac{1}{R} + \epsilon\right)^k \cdot B\left(\frac{1}{R} + \epsilon\right)^{n-k}$$

$$= AB\left(\frac{1}{R} + \epsilon\right)^n (n+1)$$

$$|C|^{1/n} \leq A^{1/n} B^{1/n} (n+1)^{1/n} \cdot \left(\frac{1}{R}+\epsilon\right)$$

$$\Rightarrow \limsup_{n\to\infty} |C_n|^{1/n} \leq \left(\frac{1}{R}+\epsilon\right)$$

$\epsilon > 0$ arbitrary, $\quad < \frac{1}{R}$.

Compare the coef. of

$C(S)$, & $A(S) B(S)$ on $|s| < R$.

$C_n \leftarrow \quad \left(\sum a_k s^k\right)\left(\sum b_k s^k\right) \Rightarrow$ cof s^n $\sum_0^n a_k b_{n-k}$

Ex:- $a_0=0,\ a_1=1,$ For $k\geq 2,$

$$a_k = \underline{a_0 k + a_1(k-1) + \cdots + a_{k-1}}$$

$$= \sum_{j=0}^{k} b_j a_{k-j}$$

Find a_n.

$b_n = n$.

$\downarrow$ Conv. of $\{b_k\}$ and $\{a_k\}$

$$c_n = \sum_{k=0}^{n} a_k b_{n-k}$$

$c_n = a_n \ \forall\, n \geq 2$

$c_0 = b_0 a_0 = 0$

$c_1 = b_0 a_1 + b_1 a_0 = 0$

$$C(s) = \sum_{n=0}^{\infty} c_n s^n$$

$$A(s)B(s)$$

$$= \sum_{n=2}^{\infty} c_n s^n = \sum_{n=2}^{\infty} a_n s^n$$

$$= A(s) - a_1 s$$

$$a_1 \to 1$$

$$= A(s) - s$$

$$A(s)B(s) = A(s) - s.$$

$$\Rightarrow A(s)(1 - B(s)) = s.$$

$$\Rightarrow A(s) = \frac{s}{1 - B(s)}$$

$P_X(s) P_Y(s) = P_{X+Y}(s)$

$X \sim Ber(p)$, $P_X(s) = q + ps$

$Y \sim Bin(n, p)$, $P_Y(s) = (P_X(s))^n$

$= (q + ps)^n$

$\{X_i\}$ iid gen fn

$P(s)$, N is another indept r.v.

having gen fn $G(s)$.

Define the random sum

$$S_N = \begin{cases} 0 & \text{if } N=0 \\ \sum_{i=1}^{n} X_i & \text{if } N=n \geq 1. \end{cases}$$

$$E\left(s^{S_N}\right) = E\left(s^{S_N} \sum_{n=0}^{\infty} I_{(N=n)} \right)$$

$$= \sum_{n=0}^{\infty} E\left(s^{S_N} \cdot I_{N=n}\right) = E\left(I_{N=0}\right) + \sum_{n=1}^{\infty} E\left(s^{X_1+\cdots+X_n} I_{N=n}\right)$$

$$= P(N=0) + \sum_{n=1}^{\infty} E\left(s^{X_1+\cdots+X_n}\right) E\left(I_{N=n}\right)$$

$$= P(N=0) + \sum_{n=1}^{\infty} \left(P(s)\right)^n P(N=n)$$

$$= G(P(s))$$

$$\boxed{P_{S_N}(s) = G(P(s))}$$

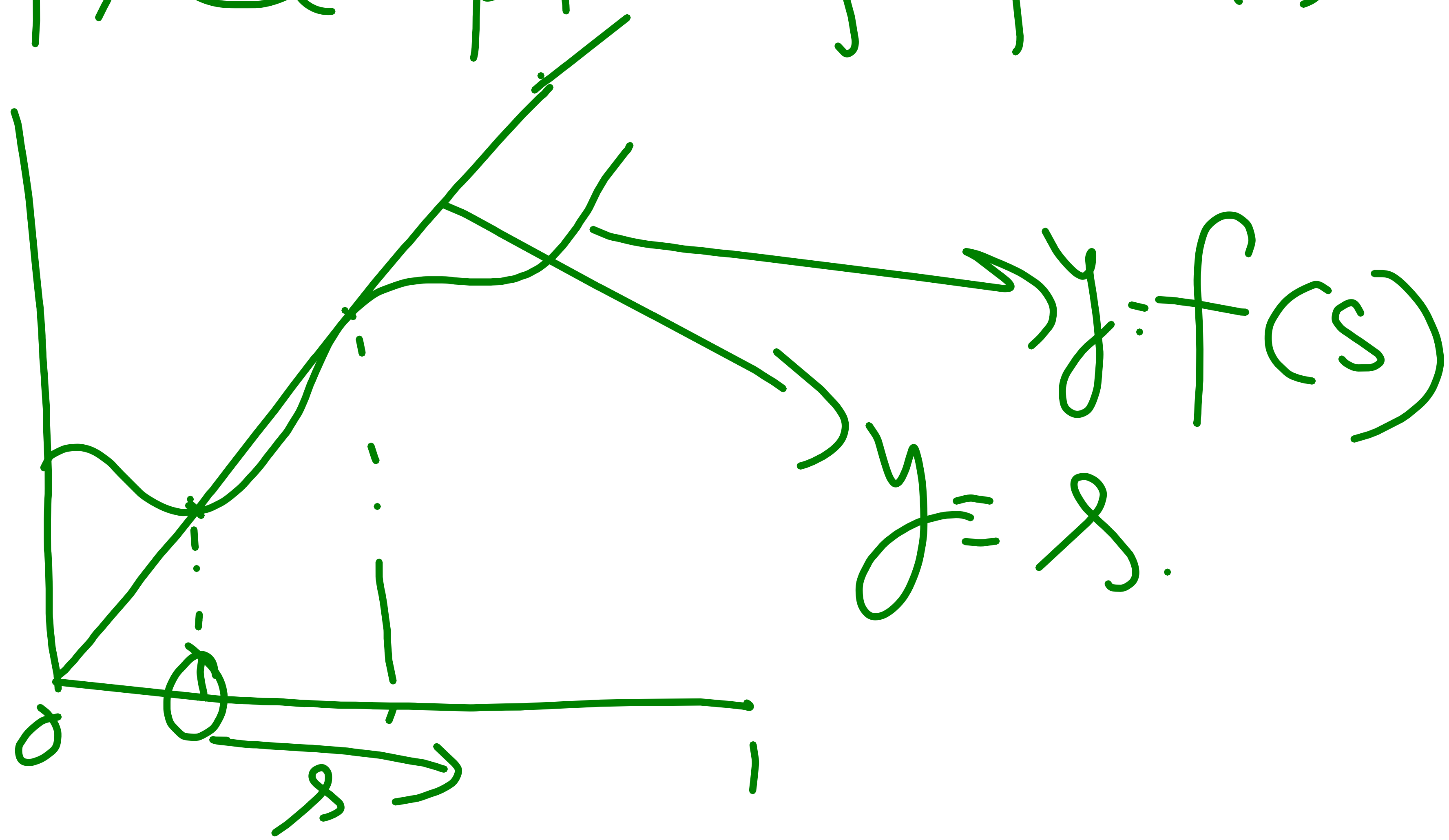
$$P'_{S_N}(s) = G'(P(s))\,P'(s)$$

Use $P(1)=1$

let $s \uparrow 1$, $E(S_N) = E(N)E(X_1)$ → Wald's eqn

Fixed pt. of pgfs in [0,1].

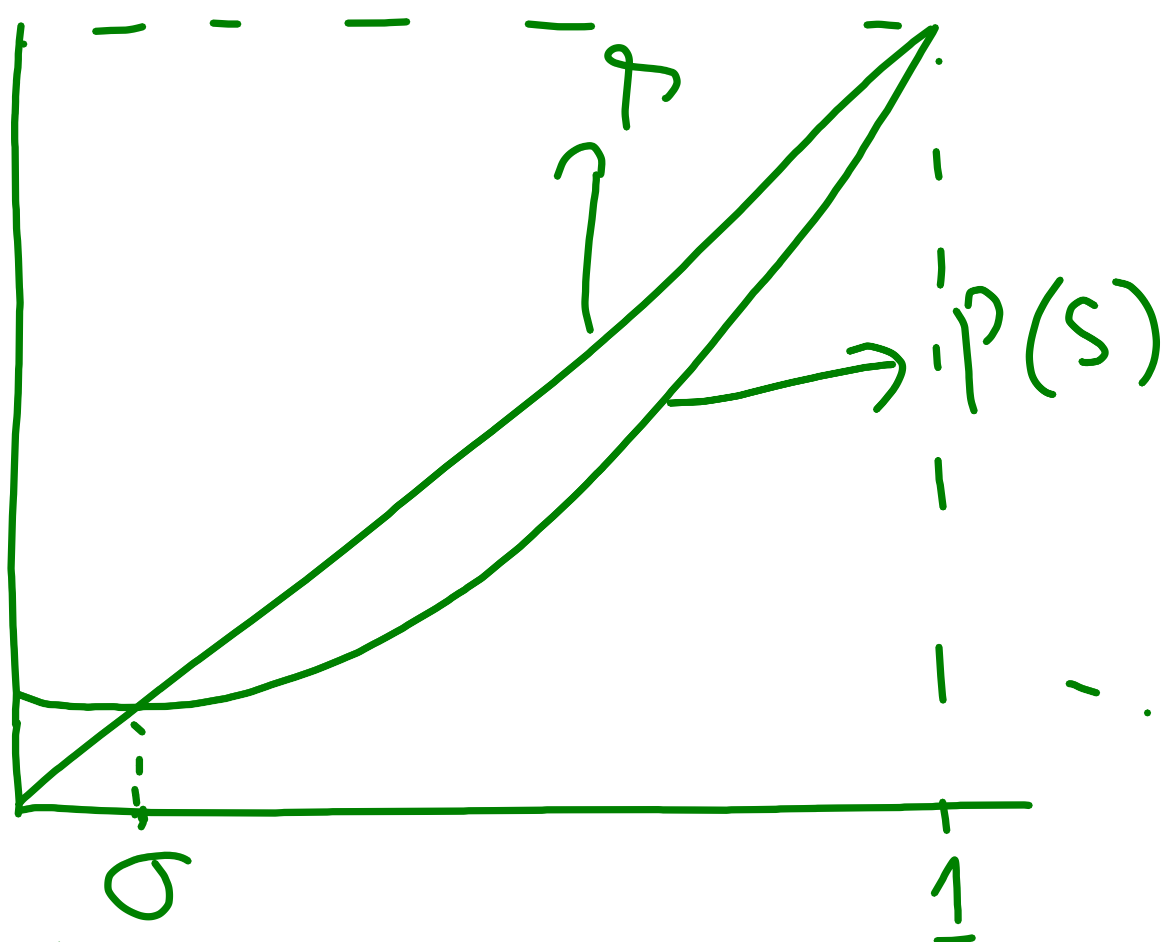
$y=f(x)$, a pt. x is called a fixed pt of f if $f(x)=x$.

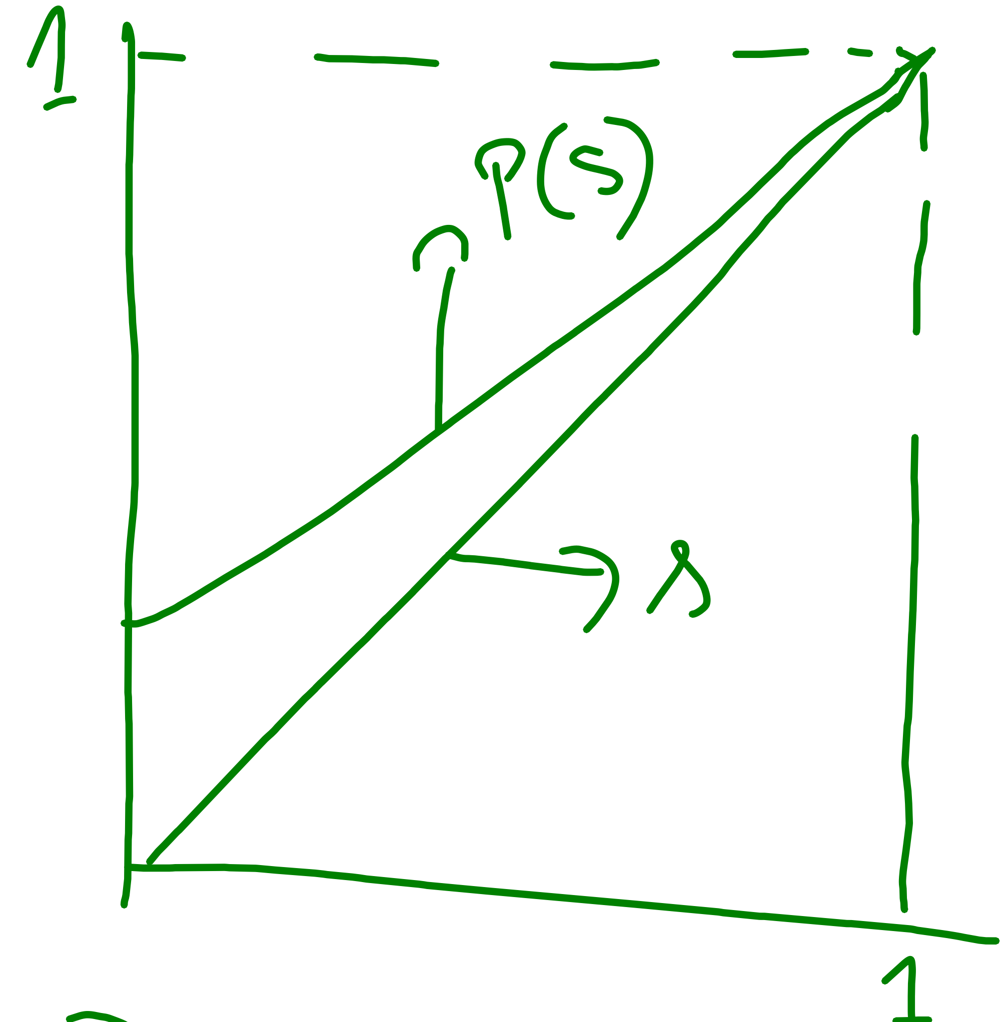


Exclude 1

$p_1 = 1.$

$p_1 < 1$





$P(s)$ is convex on $[0,1]$

$$f(a\lambda+(1-\lambda)b) \le \lambda f(a)+(1-\lambda)f(b)$$
$$\lambda\in[0,1]$$

$P(1)=1. \Rightarrow 1$ is a fixed pt.

Trivial Case

$p_1 = 1.$

Case 1 : $P(s) > s$ on $[0,1)$

$\Rightarrow 1 - P(s) < 1 - s$

$f(a) - f(b)$

$= f'(c)(a-b)$

for some $c \in (a,b)$

$\Rightarrow \dfrac{1-P(s)}{1-s} < 1$. for $[0,1)$

$\Rightarrow P'(\sigma_s) < 1$ for $\sigma_s \in (s,1)$

Letting $s \uparrow 1$, $\sigma_s \uparrow 1$.

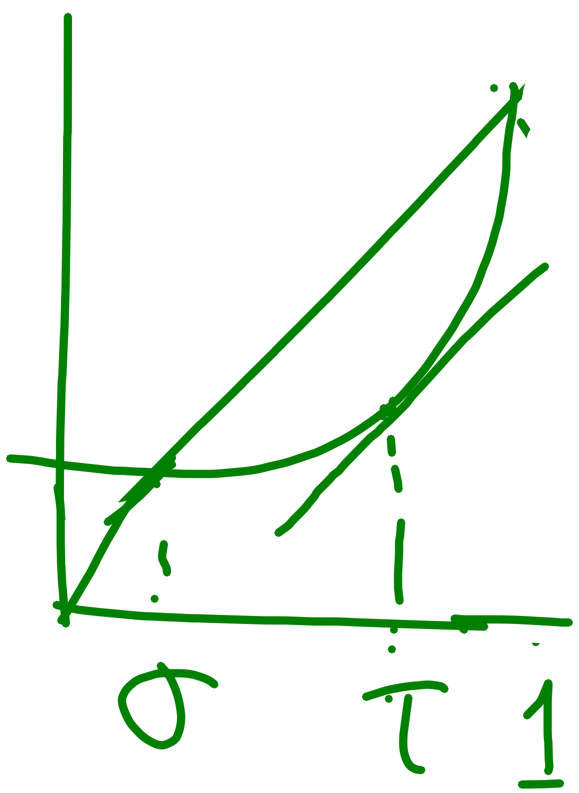
$\Rightarrow \lim_{s \uparrow 1} P'(\sigma_s) = \lim_{s \uparrow 1} P'(1) = E(X)$

≤ 1.

Case 2 : If $p_0 = 0$, $P(0) = p_0 = 0$

$\Rightarrow 0$ is a fixed pt.

$\exists$ a fixed pt $\sigma \in [0,1)$



$p_0 > 0$.

For some $\sigma \in (0,1)$,

$P(\sigma) = \sigma \Rightarrow 1 - P(\sigma) = 1 - \sigma$

$\Rightarrow \dfrac{1 - P(\sigma)}{1-\sigma} = 1 \Rightarrow P'(\tau) = 1.$

$p_1 < 1$, then $p_k > 0$ for some $k \geq 2$.

$$E(X) = \sum_{k=1}^{\infty} k p_k$$

$$= p_1 + \sum_{j=2}^{\infty} p_j \rightarrow 1.$$

$$> 1 + \sum_{j \geq 2} (j-1) p_j > 0$$

Need to show $P'(s)$ is strictly incr. If not $P'(s) = p_1$

$\Rightarrow P(s) = p_0 + p_1 s$ with $p_k = 0$ $\forall\, k \geq 2$.

This implies that two lines, $y = s$, $y = p_0 + p_1 s$ have exactly two intersection, not pos. Hence P' is st. incr. $P'(1) > P'(\sigma) = 1$

Thm For $p_1 < 1$, $P(s)$ has exactly 1 fixed pt iff only if $E(X) \leq 1$.